

PRANAV KAKDE

AI/ML & GenAI Engineer | B.E. IT '26

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EDUCATION

B.E. in Information Technology (2022 - 2026)

Pimpri Chinchwad College of Engineering and Research (PCCOER), Pune | **CGPA: 8.7 / 10**

TECHNICAL SKILLS

AI / ML & GenAI: LangChain, LangGraph, LLM Agents, RAG, RAGAs, Fine-tuning (LoRA), OpenCV, Grad-CAM, CNN-LSTM

Frameworks & Tools: Python, PyTorch, TensorFlow, Keras, Scikit-Learn, FastAPI, Streamlit, Hugging Face, OpenAI APIs

Vector Databases & Cloud: ChromaDB, FAISS, Pinecone, AWS (S3, Glue, Redshift), SEC EDGAR, yFinance

Databases & Infrastructure: PostgreSQL, Redis, Docker, Prometheus, Grafana, Git, GitHub Actions, Linux

FEATURED AI/ML & GENAI PROJECTS

TechSight – Agentic Financial Research Analyst (LangChain, LangGraph, GPT-4o-mini, AWS S3, ChromaDB, RAGAs)

- Built a multi-agent LangGraph workflow reducing financial report analysis time from hours to under 30 seconds.
- Optimized vector retrieval using ChromaDB & AWS S3 with LoRA fine-tuning, achieving 94% retrieval accuracy and near-zero hallucination rate evaluated via RAGAs.
- Delivered interactive research interface allowing custom query execution over complex financial SEC 10-K filings.

Unified LLM Gateway – Production-Grade AI Proxy Service (FastAPI, Redis, PostgreSQL, Prometheus, Grafana, Docker)

- Implemented two-layer caching (exact SHA-256 and semantic cosine similarity threshold 0.92 in Redis), reducing API latency by up to 85% for repeat queries.
- Built automatic provider failover and budget-enforcing token cost metering across OpenAI, Anthropic, and Groq endpoints.
- Exposed Prometheus metrics and Grafana dashboards for real-time throughput, error rate, and token spending analytics.

MedRAG – Modular Medical Document Processing & Info Service (LangChain, OpenAI APIs, AWS Glue, ChromaDB, SQL, OCR)

- Built an end-to-end medical document ingestion pipeline supporting PDF/DOCX/TXT files using OCR and hybrid retrieval (FAISS/ChromaDB).
- Enforced strict context-constrained prompt boundaries, eliminating external knowledge hallucination on medical queries.
- Integrated AWS Glue and SQL data transformation pipelines to structure unstructured clinical data.

FinMCP – MCP Server for Real-Time Financial Intelligence (LangChain, FastAPI, Amazon S3, Redshift, yFinance)

- Developed an MCP-compliant tool server integrating yFinance and SEC EDGAR APIs for agentic market intelligence.
- Connected Amazon S3 and Redshift data warehouses to serve historical stock metrics and SEC 10-K/10-Q financial statements.

Violence Detection with Explainable AI (Python, CNN-LSTM, Grad-CAM, OpenCV, Telegram API)

- Achieved 92.4% detection accuracy on surveillance video clips using MobileNetV2 spatial extractor + LSTM temporal modeling.
- Integrated Grad-CAM explainable AI heatmaps to visually highlight regions triggering high-confidence predictions.
- Implemented real-time video stream ingestion via OpenCV with automated Telegram bot alert notifications.

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Oracle Cloud Infrastructure (OCI) AI Foundations Associate, Generative AI & LLM Engineering Certification.

Achievements: 2nd Place at Concept Carnival 2023 for Harvest Hub; 200+ LeetCode problems solved.